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Estimating Depth of Anesthesia for Drug Infusion Using Human EEG Signals

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Abstract

General anaesthesia is a medically induced unconscious state or coma resulting from the use of anaesthetic agents on patients. It guarantees a safe and painless surgical operation for patients. However, there are numerous evidences that the patients may suffer from postoperative side effects. A limited number of solutions are currently available to deal with these issues. Therefore, optimal quantity of anaesthesia is a must to reduce the inverse effects of under-dosing or overdosing. EEG signals are, at present, most widely used signals for estimating Depth of Anaesthesia (DoA). The aim is to estimate the DoA based on the Sample Entropy of the EEG signals using Intrinsic Mode Functions (IMFs) which were extracted using Empirical Mode Decomposition (EMD) techniques. These Sample Entropy values were used along with the recorded BIS values in Machine Learning algorithms such as Artificial Neural Network (ANN) and Random Forest Regressor. In this paper, we have proposed and implemented a method for estimation of DoA using human EEG signals. We have filtered the raw EEG data using EMD and MEMD methods. And then calculated the SampEn value of 5s segments of that filtered data. Then applied different ML models to predict the values based on BIS data as target. We have taken RMSE and MAE as metrics to evaluate the models. The Random Forest Regressor model gives the results with a Root Mean Squared Error (RMSE) of 11.73 and the Mean Absolute Error (MAE) of 5.75 as the lowest error amongst all models.

Keywords: *EEG, DoA, IMFs, EMD, MEMD, BIS, Sample Entropy, ANN, Random Forest Regressor.*

1. Introduction

In the recent years, with the growth of technology, the healthcare industry has also evolved with a rapid rate. With the continuous growth in healthcare, the doctors have discovered and developed new methods as well upgraded the existing traditional methods of surgery. This includes using new hardware devices to monitor patients' vital signs instead of doing it manually.

General anaesthesia (GA) is a medically induced unconscious state which is reversible with human interaction. GA plays a very important role during surgery for patients' loss of consciousness (LoC). It helps in a safe and painless surgery. Traditionally, the anaesthetist monitors patients' vital signs such as blood pressure, heart rate and oxygen levels manually during the surgery to determine the patients' level of unconsciousness. This is a very tedious and repetitive process for anaesthetists. This method is prone to human errors as well which, in this case, may lead to very undesirable outcomes.

Anaesthetics generally show effect by decreasing the excitatory cells' activity or by increasing the inhibitor cells' activity. By this, we can conclude that during the unconscious state induced by anaesthetic agents, the communication between patients' neural network is disrupted to some extent. This disruption of communication affects the processing of information done by patients' brain. It can be inferred that the level of unconsciousness is related to the activity of brain. Thus, by monitoring the brain activity during surgery, can help anaesthetists to determine the patients' level of unconsciousness. However, monitoring brain activity during surgery is not preferred by anaesthetists because any reliable markers are not yet found.

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Brain processes information using electrical signals. These electrical signals are transmitted and received by neurons in the neural network and then sent to brain via electrical pulses. Electroencephalogram (EEG) shows the electrical activity of the brain. As we know that the brain activity is affected by anaesthetic agents, EEG signals can be used to know the extent of that affect and the level of unconsciousness can be determined by these EEG signals. In this study, these EEG signals are used for the estimation of DoA.

2. Background

General Anaesthesia (GA) is a very important part of modern-day surgical procedures. It helps patients to undergo surgery without worrying about any kind of pain. It usually results in the calm and peaceful state of patient before and after surgery. This also helps doctors to perform surgery without worrying about patients' mental state.Finding reliable markers in EEG signals for DoA estimation during surgeries has been the focus of many researches attempts over past few decades. Researchers have proved that the pattern EEG signals changes after administering anaesthesia to the person [3]. This establishes the very foundation of various commercially available products, which uses the EEG signals to provide DoA indices.

After establishing the fact that EEG pattern varies with the induction of anaesthesia, the researchers have tried to come up with a method to estimate DoA using these altered EEG patterns. A research indicates that EEG entropy measurements play an important role in anaesthesia [1]. Another research indicates that some anaesthetic agents induces more complex EEG patterns [2].

Entropy shows the randomness of a data, which in this case can be considered as the measure of complexity of the signals. There are various entropy methods to measure the entropy such as sample entropy (SampEn), approximate entropy (ApEn), permutation entropy (PE) etc. Study [7] shows that different entropies evaluates different properties. In a study, some researchers have compared twelve such entropy measures and found that Sample Entropy performed best to discriminate burst suppression pattern (BSP) [1]. BSP is an EEG signal pattern with alternating high-electrical-potential brain activity and no brain activity, which can be commonly observed in EEG pattern during anaesthesia.

Researchers also claim that the correlation coefficient between BIS and SampEn of the EEG signals are more than 0.8 [6]. Another research [10] also shows promising results in this regard with the correlation coefficient of more than 0.7. Bispectral Index (BIS) monitor is most widely used EEG based instrument to measure DoA during surgical operations. The algorithm behind this is proprietary and not disclosed to others.

George J. A. Jiang et al [4] implemented a model using SampEn of EEG signals and an ANN model to determine the patients' consciousness level. This model calculates the SampEn of EEG signals and use it as input for an ANN with BIS as target value. While another research has applied random forest algorithm combining it with ReliefF feature selection [5]. It also shows that random forest algorithm has a better accuracy compared to other machine learning algorithms.

Some researchers have focused on classifying the state of patients' consciousness into categories rather than determining a numerical value for DoA. This kind of classification show much better results compared to numerical estimation. One of such study [6], have tried to classify the state into four categories, namely deep, moderate, mild and awake. They have applied weighted k-nearest neighbour (WKNN) for the classification.

Currently, there are many EEG based indices are available for numerical determination the DoA [7]. However, the algorithms behind these indices are yet not fully reliable as they partially characterise EEG features [8].

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3. Existing System

Bispectral index is one of the many and most widely used technologies to measure depth of anaesthesia. It uses the EEG signals from patient and provides a dimensionless number between 0 and 100 as an indicator of patient's DoA level as shown in *Table 1*.

Table 1: Relationship between BIS value and depth of anaesthesia

BIS value	Depth of sedation
0	Flat line EEG
0-20	Cortical silence
40-60	Deep hypnotic state
60-80	Light moderate sedation
80-100	Awake state

BIS monitors are commercially available for use. The algorithm behind BIS monitor is proprietary and not disclosed. Even after more than two decades from its approval by US FDA, BIS index is still a controversial topic for many researchers. Some studies show that BIS reduces the chances of intraoperative awareness [11], but it could not be confirmed by some very large studies [11].Although there are other similar indices available to measure DoA levels, none of the currently available indices are perfect for DoA estimation. BIS is the most accurate of all, but still not hundred percent accurate. Also, these indices mostly have proprietary algorithms behind them.

4. Proposed Model

The proposed model shown in *figure 1*,uses the EEG signals of human brain to determine the depth of anaesthesia by calculating the sample entropy values. There is enough evidence that the sample entropy of EEG signal is correlates to the DoA [6]. Here, we are using these sample entropy values to determine DoA using machine learning algorithms.

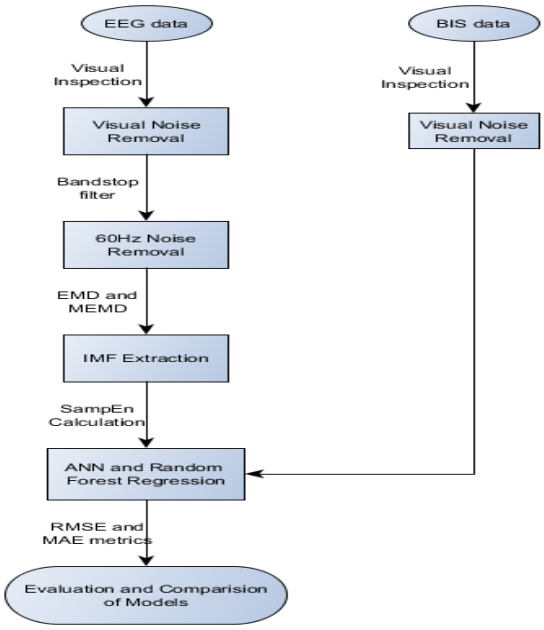

Figure 1: Flow Chart

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The data for this study was acquired from Ma, L..(2017). EEG and BIS raw data.Figureshare.[13]. "This data was recorded with the ethical approval granted by the Research Ethics Committee of the National Taiwan University Hospital (NTUH) (No: 201302078RINC). Written informed consent forms were obtained from all the patients who were involved. In this study, patients who received regular surgery under GA were recruited from NTUH (i.e., no patients underwent high risk operations related to brain, heart, lung, etc.). Those who had alcohol or smoking habits, or had other medical illness that affected the data recording, were excluded. Finally, 24 patients (ASA I or II, age (yr): 44.5 ± 12.9, height (cm): 164.2 ± 7.1, weight (kg): 63.4 ± 14.8, and BMI (kg/m2): 23.4 ± 4.2, gender: 14 females/10 males) were eligible for this study. Operations last 126.4 ± 72.9 min from anaesthetic medical recording forms." [10]

The patients' clinical information was provided to the anaesthetist to develop an anaesthetic plan for patient. "Prior to the surgery, patients were required to eat nothing for at least 8 h. All the mandatory procedures were conducted such as routine and emergency drug check. Then each patient received an appropriate volume of anaesthetic agents for the regular operations. In the surgeries, intravenous propofol was used to induce unconsciousness along with some other compounds such as muscle relaxants and analgesic. Propofol is a GABA-receptor dependent drug, which provides a rapid anaesthetic effect within a few minutes. When patients lost consciousness (i.e., no response to verbal commands or stimulus), the next stage was attained. For the maintenance period, either propofol was titrated for short-term surgeries or gas anaesthetics (desflurane or sevoflurane) together with air and oxygen in order to maintain unconsciousness. Practically, the concentration levels were adjusted within 1–1.5 minimal alveolar concentration. The anaesthetic gases were delivered by a medical mask. When approaching the end of a surgery, additional drugs are administrated to comfort the patients like vagostin, atropine, morphine, etc. To maximize the universality of the data, drugs were not specified. Thus, patients were often injected with a combination of drugs used in ordinary operations. General anaesthesia was performed safely through all stages by monitoring the physiological signals such as EEG, ECG, photoplethysmography (PPG) and by examining the intermittent vital signs of blood pressure (BP), heart rate (HR), pulse rate (PR), pulse oximeter oxygen saturation (SpO2), and so on. These measurements were shown on the physiological monitor MP60 (Intellivue; Philips, Foster City, CA, USA). If any observed signal changed irregularly, doctors will adjust the intraoperative standard anaesthesia machine accordingly. The surgical procedure and anaesthetic management details were acquired from the anaesthetic recording sheets by the hospital staff." [10].

"Prior to EEG recording, a conductive paste was used to improve the contact between the frontal scalp skin and EEG BISTM Quatro Sensor (Aspect Medical Systems, Newton, MA, US) in order to reduce the impedance under 5 kΩ. The EEG sensor was connected to the MP60 machine via the standard factory connection cable. Raw continuous EEG waveform data; including the aforementioned discrete routine vital signals; were recorded on a laptop through the serial RS-232 port utilising software developed in Borland C++ Builder 6 (C++ version 6; Borland Company, Austin, TX, USA). The EEG data were acquired at a sampling rate of 128 Hz. The intermittent BIS value was obtained every 5 s. Other intermittent vital signs such as, HR, PR, BP, and SPO2 were also recorded every 5 s using specific sensors with MP60 (e.g., pulse oximetry, ECG sensor and blood pressure cuff). The recording began 5 min ahead of the onset of induction, at which time patients were fully awake, and it terminated when patients began to respond to doctors either by voice or movement." [10]. The data is recorded as mentioned by Liu et al [10]. We have used this recorded raw EEG and BIS data for this study as it is. We do not take any credit for recording of this data. The source is cited with proper format [13].

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Steps in making the model

Visual Inspection of data:

The raw EEG data was visually inspected to check if the data is adequate or not. While inspecting the data, we found that there were some artefact segments, may be due to some electrical components or body movement. These artefact segments were removed from the data. This helped reducing outliers from the data. This can be observed by comparing *figure 2* and *figure 3*.

BIS index values range from 0 to 100, but from *figure 2*, it can be observed that there are some values of BIS that are recorded as -1. It may be caused because of some technical issues during data recording. The data points with the value of -1 were replaced by mean of 10 values on either side which were not equal to -1. Similarly, the artefacts from EEG data were removed by replacing them with the mean of 1000 values on either side. The values were replaced instead of removed to maintain the consistency between EEG and BIS data as these are recorded at different frequency.

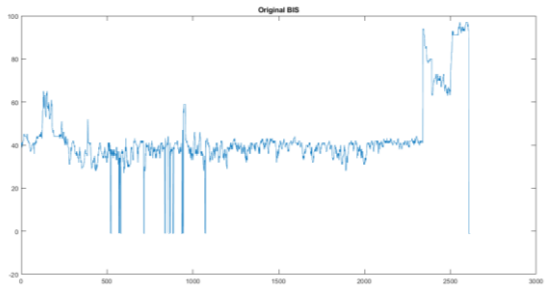

Figure 2: Recorded BIS data

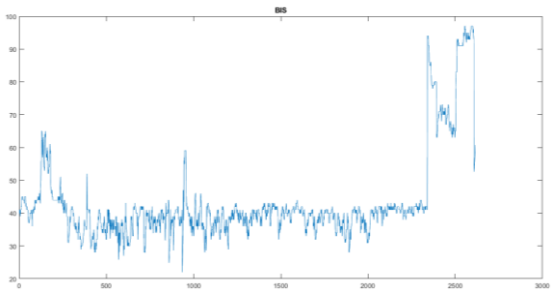

Figure 3: BIS data after removing visible artefacts

60Hz Line Noise filter:

50 Hz or 60Hz noise mainly occur due to ambient current in any building with electrical wiring [14]. A Butterworth Band stop filter was created to remove the 60Hz noise from the signal. Band stop filtersare used to screen out a certain range of frequencies. It stops the specified range of frequencies and allows the frequencies outside of that range to pass. In this case, only 60Hz frequencies were stopped and the rest were allowed to pass. By comparing *figure 4* and *figure 5*, it can be seen that the noise in the signal is greatly reduced.

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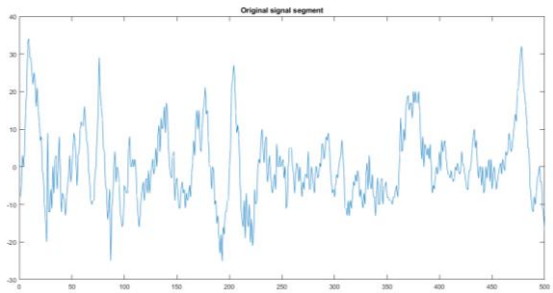

Figure 4: Raw EEG signal segment

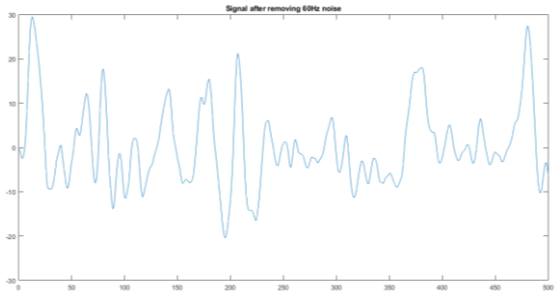

Figure 5: EEG signal segment after filtering 60Hz noise

Decomposition and Reconstruction of Signal:

Decomposition of a signal means breaking the complex signal into simpler signals, which can be easy to interpret and process. These simpler signals are known as Intrinsic Mode Functions (IMFs). An IMF is a time-varying function consisting only a single frequency. IMFs represent the oscillation modes embedded in the signal. IMFs represent the naturally derived set of basic functions of the signal.In this study, we have applied this EMD and MEMD method to decompose the EEG data and extract IMFs from it. We have dropped IMF1. We used IMF2 and IMF3 to reconstruct the signal for further processing. The reason behind dropping IMF1 is that it constitutes most of the noise from the original signal without having much useful information in it.

a) Using Empirical Mode Decomposition (EMD):

The EMD is a method which is used to decompose a complex signal into small and finite number of functions, known as Intrinsic Mode Functions. It is mainly used in decomposition of non-stationary and non-linear time series data. It decomposes the given series into IMFs without leaving the time domain.The IMFs from a signal are extracted using a method called shifting. First, an envelope for local maxima is formed by drawing cubic spline through all local maxima. Similarly, an envelope for local minima is formed by drawing cubic spline through all local minima. Then a signal, m1, is generated by taking the mean values of both envelopes.

The IMF extraction process starts from here. The first pseudo-IMF, f1, is generated by taking the difference between original signal S(t) and mean m1,

$$f_1 = S(t) - m_1$$

And then treating this obtained pseudo-IMF as new signal, again the above steps are applied and the difference is calculated,

$$f_{12} = f_1 - m_{12}$$

After shifting n times,

$$f_{1k} = m_{1k} - f_{1(k-1)}$$

where, f1k is a newly derived IMF.

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The above steps can be repeated for the residual to generate desired number of IMFs.

The original signal can be represented as

$$S(t) = \sum_{n=1}^{N} (IMF_n(\mathrm{t})) + res$$

where

- N is the number of extracted IMFs
- res is the residual.

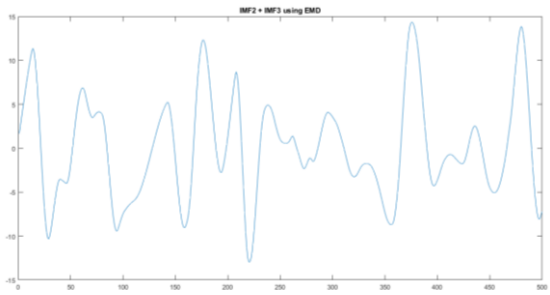

Figure 6: Reconstructed signal segment after filtering by EMD

b) Using Multivariate Empirical Mode Decomposition (MEMD):

MEMD is an improved version of previously existing EMD algorithm [15]. MEMD gives better performance compared to EMD while applying on non-linear and non-stationary data series.

The MEMD is performed using the equation

$$mean(t) = \frac{1}{l}\sum_{i=1}^{l} e^{\theta i}(t), \text{Where}$$

$e^{\theta i}(t)$is the multivariate curve

l is length.

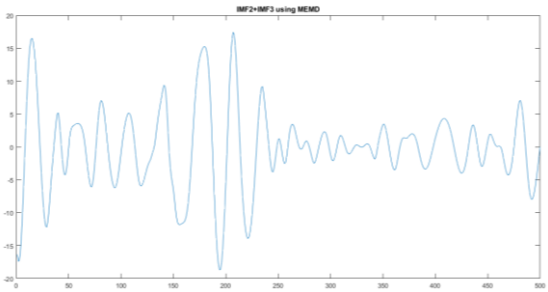

Figure 7: Reconstructed signal segment after filtering by MEMD

Sample Entropy Calculation:

Sample entropy is a measure of randomness in a data series. It can also be interpreted as the complexity of time-series signals. “The more complex the signal is the higher the value of SampEn”. And in our case, the more complex the EEG signal pattern is, the more awake the patient is. Thus, it can be said that the SampEn values of EEG signals can be directly use to interpret the patients’ level of unconsciousness.

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Assume the given time-series data with constant time interval t, as
$X = \{x_1, x_2, x_3, \ldots, x_N\}$

The steps to calculate SampEn are:

a.) Determine the length of given time-series data as N and constant time interval t.

b.) Determine the values of embedding dimension m, tolerance r.

c.) Create a temporary vector with length m as

$$X_m(i) = \{x_i, x_{i+1}, x_{i+2}, \ldots, x_{i+m-1}\}$$

d.) Create a distance function using Chebyshev distance

$$d[X_m(i), X_m(j)] \text{ where } i \neq j$$

e.) Determine A as number of pairs having $d[X_{m+1}(i), X_{m+1}(j)] < r$.

f.) Determine B as number of pairs having $d[X_m(i), X_m(j)] < r$.

g.) Calculate Sample Entropy as

$$SampEn = -\log\frac{A}{B}$$

In our study, we calculated the SampEn value of 5sec segments of the filtered EEG signal. The length was chosen as 5sec to make the SampEn values consistent with BIS values, which were originally recorded at 0.2Hz frequency. The EEG signal was recorded at 128Hz frequency and thus the SampEn of 640 values (128Hz * 5sec) were calculated.

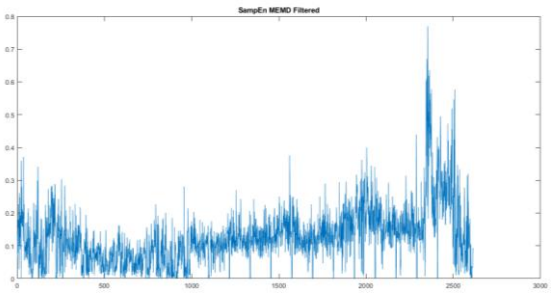

Figure 8: Sample entropy values graphafter filtering by MEMD

It can be visually interpreted from figure 3 and figure 8 that the SampEn values are directly correlated to BIS values.

Machine Learning Models:

From *figure7* and *figure 8*, we can visually interpret and say that the SampEn of EEG signals and BIS values correlates to each other. But we can only use our qualitative reasoning to do so. We, as humans, are unable to quantitatively process this huge chunk of data in short amount of time. To process this huge amount of data, we need to rely on modern computers for their high computational power. And to enable a computer to be able to do decision-making, we need to use machine learning algorithms.

We are focusing here on estimating the DoA using a numerical value between 0 to 100, instead of categorising the DoA levels in multiple categories. To establish a quantitative relationship between SampEn values and BIS values, we are using some machine learning algorithms.

a) Artificial Neural Network (ANN):

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Artificial Neural Network (ANN) are computational systems based on the working of brain. Brain consists of millions of neurons that processes the data. These neural networks 'learn' by examples rather than some predefined formulae or rules.The ANN consists of multiple layers of neurons. Each layer takes data from previous layer as input and processes it. Based on the previous error, it penalises and updates the estimators.

There are various types of neural networks which are used based on the complexity of problem and the use cases. Here, the network type was taken as 'feed-forward backpropagation'. The hidden layer structure was taken as 10-8-9-1.

b) Random Forest Regressor:

Random forest is an ensemble learning method for classification and regression of data. Ensemble learning method uses multiple machine learning models and combines their results. It is done to improve the overall accuracy of the model. It gives greater accuracy than any individual model could provide.Random forest uses the Bagging (Bootstrap Aggregation) technique. In these techniques, multiple decision trees were formed and trained using separate subset of original data. The formation of subsets was done on original data with replacement. Here, all decision trees are trained separately and then an average is taken from all the predictions from all different decision trees. It helps in increasing accuracy of overall model.The algorithms were applied on three different datasets, i.e. original data, EMD filtered data and MEMD filtered data, to compare and find the best model.

5. Results

After applying both the algorithms on differently filtered data, and we have found that all the models performed quite well on prediction of DoA. We took RMSE and MAE values as metrics to evaluate the performance of models. We also visually inspected the graphs between original values and predicted values. These graphs are attached after each implementation of model in corresponding sections.

Table 2 shows us the comparison between different implemented model using RMSE and MAE as metrics for comparison. From T*able 2*, we can conclude that both ANN and Random Forest Regressor has most accuracy while using MEMD filtered data. And the Random Forest Regressor out-performs ANN.The model with least error is generated by applying random forest regressor on MEMD filtered data and their outputs are represented in figure 9 and figure 10. The RMSE value of this model is 11.7382 and MAE value is 5.7545, which are the lowest among all models.

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Table 2: Comparison of error values for different ML models

ML Algorithm	Data	RMSE	MAE
ANN	Original	13.0609	7.5997
	EMD filtered	12.5715	7.0393
	MEMD filtered	12.1193	6.5874
Random forest regressor	Original	13.4173	6.7504
	EMD filtered	12.6995	6.4782
	MEMD filtered	11.7382	5.7545

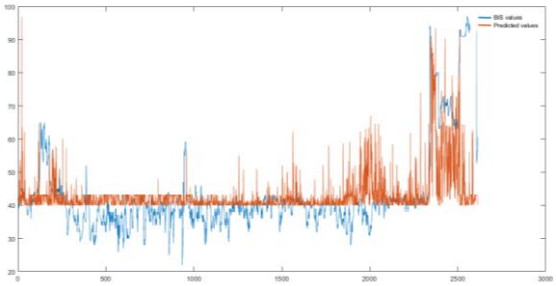

Figure 9: Graph between Original (blue) and Predicted (orange) BIS values by ANN using MEMD filtered data

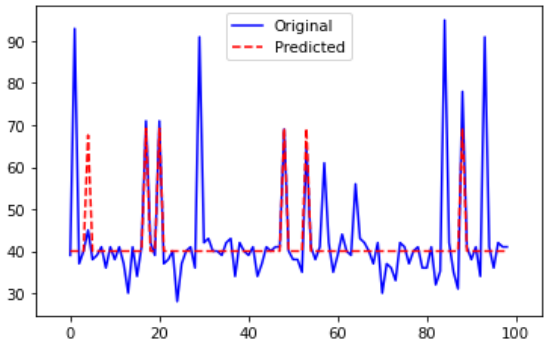

Figure 10: Graph between Original (blue) and Predicted (red) BIS values by Random Forest Regressor using MEMD filtered data

6. Conclusion

In this project, our aim was to generate various models to determine the DoA. We used different machine learning algorithms as well as different set of data. This was done to find the optimal machine learning model with best accuracy.We have found that all models were performed almost similarly. There was not much change when MEMD filtering was used instead of using the signal directly. The main increase in accuracy was observed after the removal of artefacts after visual inspection. "A typical adult human EEG signal is about 10 µV to 100 µV in amplitude when measured from the scalp and is about 10–20 mV whenmeasured from subdural electrodes" [9]. From this, we can infer that artefacts due to electrical interference play a very important role in the studies related to EEG signals.

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To omit these artefacts, one method is to use a 'Faraday Cage'. But realistically, it is not possible during surgeries with a lot of electrical equipment around. To overcome this issue, we need an improved method for recording of EEG signals. From this project, we have also determined that sample entropy of EEG signals plays a huge role in DoA estimation. But we also realised that this is not the only factor that is important for DoA estimation. There might be other factor which could be used along with sample entropy for DoA determination. But not much research is done to find the potential factors.

All models performed with almost equal accuracy. Although, through MEMD filtering, the error was reduced, we can see that the difference was very small. We can also say that the MEMD filtering was not very effective while filtering non-linear non-stationary time-series. There is yet another improved version of MEMD which can be used to filter the data. This method is known as N-A-MEMD (Noise-Assisted Multivariate Empirical Mode Decomposition) [12]. This might further improve the accuracy of machine learning models. This method was not applied in this project due to time and computational constraints.

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